Part A

Literature Review and Public Datasets

Publication 1:

Title	**Machine Learning Methods for Bank Term Deposit Subscription Prediction**
Havard Reference	Wu, X., 2026. Machine Learning Methods for Bank Term Deposit Subscription Prediction. *International Journal of Global Economics and Management,* 10(4), pp. 36-43.
Link	https://doi.org/10.62051/ijgem.v10n4.06
Modelling	The processed data are then uniformly fed into three classification models, namely Logistic Regression, Random Forest, and LightGBM, for training and testing
Preprocessing Techniques	Missing values were inspected, categorical-variable encoding, SMOTE for class imbalance and feature standardisation, correlation-based feature screening. A 70:30 train/test split, a correlation threshold of 0.35, and selection of 9 features from a total of 21 variables.
Datasets Used	The Bank Marketing dataset containing customer, marketing and economic-background variables for predicting whether a customer would subscribe to a term deposit. The study reports an approximately 8:1 class imbalance.
Key Takeaways	Explicitly consider class imbalance, and don't rely on accuracy alone, since it can be misleading when one class is much larger than the other. Tree-based boosting models such as LightGBM can capture nonlinear relationships and interactions between variables in customer-level tabular data. Using the same preprocessing and evaluation framework when comparing models helps ensure that differences in performance are attributable to the modelling approach rather than inconsistent experimental conditions.

Publication 2

Title	**Applying Machine Learning to the Development of Prediction Models for Bank Deposit Subscription**
Havard Reference	Hou, S., Cai, Z., Wu, J., Du, H. and Xie, P. 2022. Applying machine learning to the development of prediction models for bank deposit subscription. *International Journal of Business Analytics*, 9(1): 1–14. doi:10.4018/IJBAN.288514.
Link	https://doi.org/10.4018/IJBAN.288514
Modelling	Naïve Bayes, decision tree, random forest, support vector machine using an RBF kernel, and neural network models were evaluated. Random forest was used as an ensemble of decision trees. The authors also performed a sensitivity analysis to compare predictive performance.
Preprocessing Techniques	Missing and 'unknown' categorical values were investigated. 'Unknown' was retained as a category for some variables because the number of true positive cases was very small. Observations with unknown values in several other categorical variables were excluded. The resulting 38,245 observations were divided using stratified sampling into 66% training and 34% testing data.
Datasets Used	The UCI Bank Marketing dataset, which contains information from direct marketing campaigns conducted by a European bank. The study reports 45,211 observations, a binary target indicating whether the client subscribed to a term deposit, and 20 independent variables.
Key Takeaways	Stratified splitting helps preserve the class distribution between training and testing data, which matters when the target classes are imbalanced. Missing-value treatment should consider the meaning and distribution of missing values rather than automatically applying the same imputation method to every variable. Comparing several model families provides a stronger basis for selecting a suitable model than relying on only one algorithm.

Publication 3

Title	**A data-driven approach to predict the success of bank telemarketing**
Havard Reference	Moro, S., Cortez, P. and Rita, P. 2014. 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22–31. doi:10.1016/j.dss.2014.03.001.
Link	https://doi.org/10.1016/j.dss.2014.03.001
Modelling	Logistic regression, decision tree, neural network and support vector machine were used. The study applied semi-automatic feature selection, reducing 150 candidate features to 22 features. A realistic rolling-window evaluation approach was used to assess model performance on newer observations. The neural network produced the strongest reported results.
Preprocessing Techniques	Feature engineering and business-informed feature selection were applied. The study considered client, product and social/economic variables. A forward feature-selection approach was used to reduce the number of variables. The modelling evaluation used a time-aware rolling-window approach.
Datasets Used	Real Portuguese retail-bank telemarketing data collected between May 2008 and June 2013. The dataset contained 52,944 records and 150 candidate features before feature selection.
Key Takeaways	Time-aware validation is relevant when customer behaviour and marketing activity change over time, which can inform the validation strategy for STADIOEquities. Feature engineering and domain-informed feature selection can reduce a large number of variables while retaining useful predictive information. Model evaluation should reflect the intended business use case rather than relying only on a random train/test split.

Publication for Part B

Name	**Bank Marketing**
Havard Reference	Moro, S., Rita, P. and Cortez, P. (2014) *Bank Marketing* [Dataset]. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C5K306.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of Instances	**45,211 observations** in the selected **bank-full** dataset. The UCI repository also provides smaller/alternative versions: bank.csv (4,521), bank-additional-full.csv (41,188) and bank-additional.csv (4,119). For Part B, the selected dataset will be **bank-full**, containing 45,211 observations.
Number of feature columns	**16 feature/predictor columns**, excluding the binary target column y. Therefore, the selected dataset contains **17 columns in total** when the target variable is included.
Number of classes	2 classes: yes and no
Target Range	Not applicable because this is a **classification** problem.
Origin	The dataset originates from direct-marketing campaigns conducted by a Portuguese banking institution. Clients were contacted by telephone during marketing campaigns, and the target variable records whether the client subscribed to a term deposit. The UCI Machine Learning Repository provides several versions of the dataset.